AI Assignment

Part 1: Find the best AI Employee

Objective: Find 3 LLM's and test them equally to find the best one

- **List your chosen 3 LLM's.**

  1st – Cursor
  2nd – ChatGPT
  3rd – GitHub Copilot

- **Now it's time to craft your interview questions. I already crafted 2 questions for you, craft the last 3 questions and make sure they are web development related questions.**

  1st – Craft a TicTacToe application with HTML, CSS, and JS.
  2nd – Explain the previous code generated
  3rd – Improve the previous game to make it player vs AI
  4th – Make a portfolio website
  5th – Explain ways to make portfolio websites stand out

- **Report the results of the 1st LLM for each question. Simply report how accurate, clear, and efficient (ACE) each response was.**

  1st – Craft a TicTacToe application with HTML, CSS, and JS.
  The tictactoe application is accurate. The coding style is different from what I usually use/understand therefore it takes time for me to understand the code. The response is very efficient because Cursor will create all the files for me, so I don't need to copy and paste the code from the conversation to VS Code.

  2nd – **Explain the code for the following application from the 1st question.**
  The explanation is accurate and easy to navigate since it's side by side with the code. The explanation is simple and clear with focus on high level idea, rather than the individual syntax. The explanation is easy to navigate, making it very efficient.

  3rd – Improve the previous game to make it player vs AI
  The upgraded version is accurate where the AI use minmax algorithm to calculate the best choice to make. The coding style is different from my coding style, and it takes time to understand. The response is very efficient since it auto generate all the file and write the code to the file.

4th – Make a portfolio website

The website is accurate with all important sections for portfolio is included. The coding style is not like what I used to, for example, the CSS and JS is in its own folder rather than in the same directory as index.html. The coding style is hard to understand. The website looks complex and very well designed. It is not very efficient since I need to fill in all my details into the placeholder left in the code because Cursor AI does not have record of my name and my portfolio from my past conversation.

5th – Explain ways to make portfolio websites stand out

The response is accurate and logical. The explanation is very vague, with no clear suggestions on what I suppose to do.

- **Report the results of the 2nd LLM (ChatGPT) for each question. Simply report how accurate, clear, and efficient (ACE) each response was.**

1st – Craft a TicTacToe application with HTML, CSS, and JS.

The tictactoe application is accurate. The coding style is simple and clear. The response is efficient because it put the html, css and javascript in separate box so it's easier to copy to VS Code.

2nd – Explain the code for the following application from the 1st question.

The explanation is very clear and it follows my style which is explains in simple and direct english and it focus on helping me to understand it for exam/interview (without being explicitly stated).

3rd – Improve the previous game to make it player vs AI

The upgraded version is accurate where the AI use minmax algorithm to calculate the best choice to make. The coding style is clear and simple where every syntax is easy to understand and mostly self-explanatory. The response is very efficient since it only suggest me to change the javascript file from the previous response.

4th – Make a portfolio website

The portfolio website is accurate with all the necessary section is included. The coding style is very simple and clear which makes it easy to understand. However, there is too much unused space and the website looks too plain and boring. The response is efficient because it include all the details about me such as name, projects (from previous use) without being explicitly stated.

5th – Explain ways to make portfolio websites stand out.

The response is accurate and logical. The explanation is direct and easy to understand. The response is efficient and easy to understand/implement.

- **Report the results of the 3rd LLM for each question. Simply report how accurate, clear, and efficient (ACE) each response was.**

1st – Craft a TicTacToe application with HTML, CSS, and JS.
The game created is accurate. The coding style and syntax is easy to understand and almost self explanatory. The AI is effficient since it's built in inside VSCode so there is no need to copy and paste the code. All the codes and directories are automatically generated.

2nd – Explain the previous code generated
The explanation is by far the best since it explained the high level idea of the code as well as the logic behind the syntax. The copilot also highlighted the code in the explanation and make the explanation linked to the code so that I can go to the dedicated code section.

3rd – Improve the previous game to make it player vs AI
The game created is accurate. The coding style and syntax is easy to understand and almost self explanatory. The AI is effficient since it's built in inside VSCode so there is no need to copy and paste the code. All the codes and directories are automatically generated.

4th – Make a portfolio website
The portfolio is accurate with all sections for portfolio included. The coding style is simple and easy to understand. The AI is effficient since it's built in inside VSCode so there is no need to copy and paste the code. All the codes and directories are automatically generated. The website style looks plain and simple, suitable for professional use.

5th – Explain ways to make portfolio websites stand out
The response is clear and easy to understand. However, the explanation is very vague, with no clear suggestions on what I suppose to do.

- **Based on your interviews, which LLM had the best responses. Explain your answer.**

  ChatGPT because I like the coding style and the explanation is easy to understand.

Part 2: AI Guidance for unknown Topics

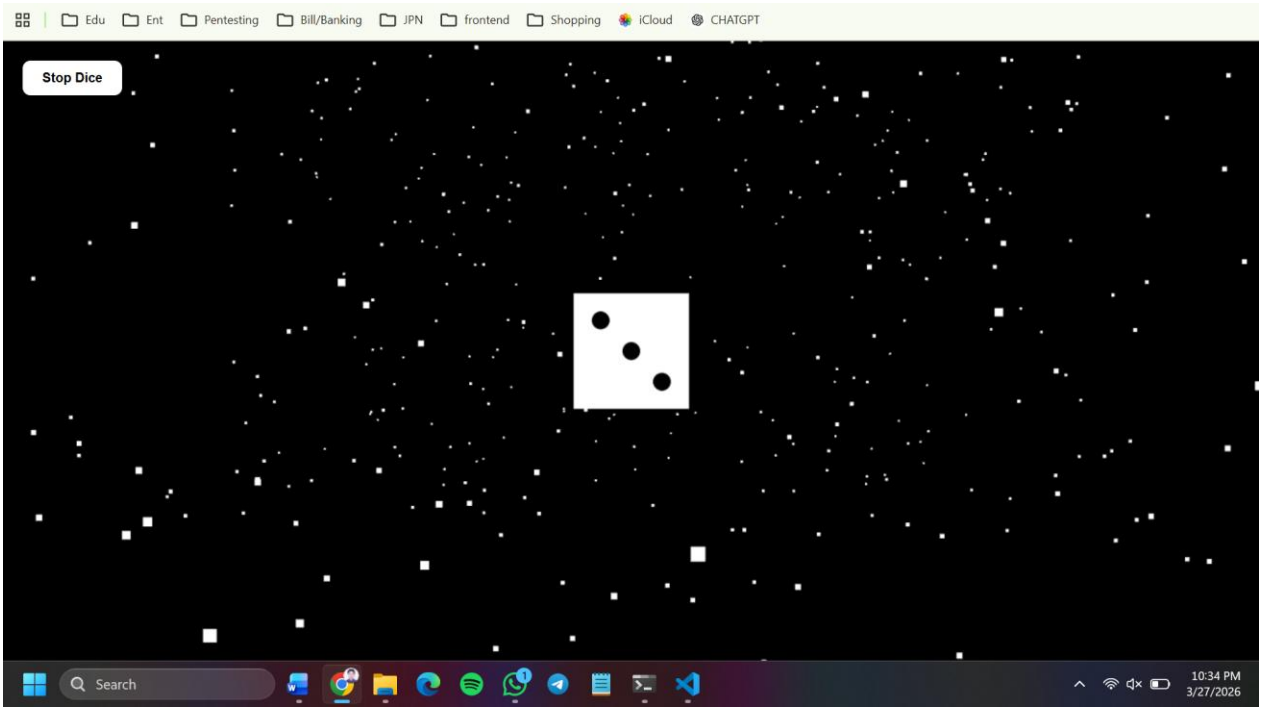
Objective: Let AI write code using Three.js library to mimic results of the following:

- **Your goal in part 2 is to create a website using AI ONLY that looks exactly similar to this website: https://roll-dice-beta.vercel.app/**
- **You have up to 10 prompts you can ask the AI. Keep in mind that you're not graded based on how similar your website looks to the given website. Just try your best to get it done in 10 prompts, if you couldn't get it done in 10 prompts you will still receive the full credit.**

Report your prompts here, then submit your github link for the website and screenshot:

Ex) "ChatGPT make me a program of a bouncing 3d cube using Three.js, HTML, CSS, JS…"

1. Create a rolling 3d cube dice animation using tree.js, html, css, js that include a stop button, bouncing animation, snowy background
2. make the dice include numbers, and the bouncing animation is throughout the screen and bouncing back and forth when it hit the edge of the screen
3. Generate the whole code to make the dice stop completely when the stop button is pressed and make it hit the edge of the screen to bounce
4. update the whole to make the stop button make the dice stop, and show the dice face at the middle of the screen, then when user press anywhere else, the dice start moving again

Github Link: https://github.com/ayishh/CMPSC263-Project2.git

Screenshot: